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PROJECTS

Munchly | *React Native, Expo, Next.js, TypeScript, Supabase, Stripe Connect, Postgres*

Live at munchly.sg

- Shipped a two-sided marketplace across three surfaces: an Expo iOS/Android app, a Next.js 16 web + seller dashboard, and an Express/Stripe Connect microservice that handles all transaction processing and per-seller payouts.
- Built a server-authoritative payments pipeline: client posts only listing IDs; server resolves prices, builds platform-charge Checkout Sessions, and issues per-seller Transfers idempotently on each verified checkout completion webhook, keyed by session ID to prevent double-writes.
- Architected an RLS-forced multi-tenant Postgres schema with Row Level Security on every public table and storage paths scoped to the authenticated user's identifier; service-role usage confined to the Stripe server.
- Implemented per-order realtime chat and order status via Supabase Realtime, with read-timestamp-driven unread badges and multi-seller carts that split one checkout into per-seller orders.
- Modeled money as integer cents end-to-end; GST, commission, and per-seller allocations computed once server-side and replayed from session metadata at the Transfer step.
- Shipped native maps per-platform (Apple Maps on iOS, Google Maps on Android), Sign in with Apple + Google OAuth, and a Vercel + Railway + EAS deploy pipeline with OTA updates.

SnapRoom | *React, TypeScript, Three.js, React Three Fiber, World Labs API, Fal AI, Expo*

2nd Place Winner, Ralphthon @SG 2026

- Built SnapRoom, a platform that reconstructs a single floor plan, sketch, or room photo into a fully walkable 3D world, orchestrating environment generation, 3D object meshes, and ambient sound from one source image set.
- Generated explorable environments as 3D Gaussian splats through the World Labs API and rendered them in real time with the Spark renderer over Three.js and React Three Fiber, serving multi-resolution splat assets so phones and laptops each load device-appropriate detail.
- Engineered an AI semantic-label layer that captures the live viewport, runs Gemini Vision object detection, and raycasts each detection back into 3D to anchor floating labels, with importance ranking that keeps only the most relevant labels on screen.
- Built a companion Expo and React Native AR viewer that opens a room from a scanned QR handoff, streams it into a WebView, and drives the in-scene camera from the phone's fused motion sensors for a walk-around experience.
- Generated furniture meshes and ambient plus per-object sound effects through Fal AI, layering real 3D meshes over the splat environment with switchable lit, scene-only, object-only, and wireframe render modes.
- Architected a disk-first asset pipeline with an indexed manifest scheme for worlds, objects, and audio, plus a Vite dev-server plugin that runs multi-minute generations detached and hot-reloads the viewer as assets land.

HACKATHONS

Ralphthon @SG 2026 Hackathon: *SnapRoom* (2nd Place Winner)

17 May 2026

React, TypeScript, Three.js, React Three Fiber, Gaussian Splatting, World Labs API, Fal AI, Expo

Singapore

- Placed 2nd building SnapRoom, a platform that turns a single room image into a walkable 3D world using Gaussian-splat environment generation, AI-generated object meshes, an AI semantic-label layer, and a companion mobile AR viewer.

AI Engineer Singapore 2026 Hackathon: *DataForge*

9 May 2026

Next.js, TypeScript, Convex, OpenAI Responses API (GPT-5.5), Fal AI, Zod, Tailwind

Singapore

- Shipped an end-to-end AI engineering prototype (agents, retrieval, structured-output evaluation) alongside OpenAI, Cursor, Vercel, Google DeepMind, and ElevenLabs.

GrabMaps 2026 Hackathon: *Personality Map*

24 Apr 2026

React + Vite + TS, Vercel Serverless BFF, Grab Maps, OpenAI, ElevenLabs

Singapore

- Built a tap-a-place experience on Grab Maps where archetype characters (Italian chef, mall guard, etc.) deliver review-flavored monologues in ElevenLabs voices and live-chat via an OpenAI chat model.
- Designed a Vercel Serverless BFF that brokers every external call (Grab Maps style/POI/Directions, Google Places reviews, OpenAI, ElevenLabs TTS) so no API keys reach the browser; map tiles proxied through a dedicated backend route.

- Engineered a fail-fast orchestrator with per-call abortable timeouts (8s LLM, 12s TTS, 5s reviews), a Zustand client cache, and an in-memory backend cache keyed by place identifier; vendored a minimal local Grab Maps package wrapping MapLibre when the public registry copy wasn't available.

NUS Hack&Roll 2026: *Reely*

17 Jan 2026

React + TS, Node/Express, MongoDB, Gemini Vision, Supabase, GCP Cloud Run

Singapore

- Built a GeoGuessr-style game that analyses uploaded images and videos with the Google Gemini Vision API to identify real-world coordinates, then drops players onto an interactive Google Maps + Street View board to guess the location with distance-based scoring and a leaderboard.
- Implemented video scene detection via FFmpeg to extract and locate multiple distinct scenes from a single clip, with Mongoose schemas modeling uploads, users, folders/collections, and game state on MongoDB Atlas.
- Shipped production-ready: Supabase Google OAuth, Dockerised frontend (Nginx) and Node/Express backend, deployed to GCP Cloud Run with secrets pulled from GCP Secret Manager.

Google Gemini 2026 Hackathon: *Urban Incident Review*

10 Jan 2026

Next.js 14, TypeScript, Express, Gemini 3 Pro, Gemini Files API

Singapore

- Built an AI video analysis platform that takes natural-language queries against uploaded surveillance footage and returns structured findings: precise timestamps, confidence scores (0.0–1.0), spatial context, and visual-evidence descriptions, with zero traditional CV infrastructure (no YOLO/DeepSORT).
- Used the Gemini Files API for upload storage and file-state transitions and Gemini 3 Pro for native long-video understanding (with Gemini 1.5 Pro fallback); supports MP4/MOV/AVI/MKV/WebM up to 2GB with async processing and status polling.
- Shipped a strict-mode TypeScript stack with a Next.js 14 + Tailwind frontend and an Express backend (Multer + Sharp), exposing a REST API returning JSON findings keyed by timestamps in seconds.

EXPERIENCE

Software Engineer Intern

May 2026 – Present

Twiss

Singapore

- Built WatchTower, an AI operations layer for F&B businesses that ingested RTSP camera feeds and turned raw video into a queryable event stream covering queue buildup, table turnover, kitchen pickup congestion, and staff coverage, with realtime alerts and natural-language “ask-your-camera” queries.
- Designed a multimodal reasoning pipeline that fused YOLO and Vertex AI Vision detections with an OpenAI LLM summarisation layer to answer free-form operational questions and emit autonomous action recommendations across staffing, restocking, and counter allocation.
- Engineered a semantic video memory layer that compressed raw frames into structured events in Supabase/Postgres, enabling decision-simulation queries against historical footage without re-processing video.
- Shipped a tiered token/cost-optimisation layer that ran a cheap vision model for steady-state monitoring and escalated to a stronger multimodal model only on anomaly triggers (crowd spikes, blocked aisles, abandoned counters), on a FastAPI + Next.js stack with Supabase Realtime over WebSockets deployed on Cloud Run.

Software Engineer Intern

Jan. 2026 – Present

Biohackk

Singapore

- Built a full-stack mobile health-tech app with secure auth (Singpass OAuth, Google OAuth) using state, nonce, and token validation for end-to-end auth security.
- Designed Supabase + GCP backend services and data pipelines, resolving data inconsistencies and shipping scalable schemas for user metrics, streaks, leaderboards, and health tracking.
- Integrated Apple HealthKit and third-party IoT/biometric SDKs (Omron, smart scales, Visbody) for realtime ingestion, sync, and bidirectional write-back across sources.
- Shipped AI-powered nutritional analysis (OCR + LLM APIs), biometric auth, push notifications, and dynamic data visualisation dashboards.

Software Engineer Intern

May 2025 – Jul. 2025

DNDTS PTE. LTD.

Singapore

- Built and maintained a full-stack internal operating system with React (Vite, Mantine) and a Go backend on SQL + Docker.
- Shipped new features and system enhancements; optimised performance and integration across microservices and internal tools.

Vice-President

NUS Students' Computing Club

- Led 100+ members across subcommittees to organise 7 major faculty-wide events impacting 5,000+ computing students.
- Coordinated hackathons, tech talks, and workshops; streamlined workflows and mentored junior leaders for continuity.

Sep. 2024 – Present

Singapore

Business Development Executive

NUS Fintech Society

- Secured sponsorships and partnerships for large-scale fintech events; built strategic collaborations to enhance engagement and funding.

Aug. 2024 – Present

Singapore

EDUCATION

National University of Singapore

Bachelor of Computing (Honours), Computer Science

Aug. 2023 – May 2027 (Expected)

Singapore

Temasek Polytechnic

Diploma (Merit), Biomedical Engineering

Apr. 2018 – Apr. 2021

Singapore

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, HTML/CSS, C/C++, SQL (Postgres), R

Frameworks: React, React Native, Next.js, Expo, Node.js, Flask, FastAPI, JUnit

Developer Tools: Git, GitHub, Docker, Supabase, Firebase, Convex, Stripe Connect, Vercel, Railway, EAS, GCP, Sentry, VS Code

Libraries: pandas, NumPy, Matplotlib, TensorFlow, PyTorch, OpenCV, scikit-learn